Close

```
# Capstone Project Part 4 – Task 1
## Document Loading and Preprocessing (RAG Pipeline)

### Objective
- Load a document (PDF/TXT/DOCX)
- Preprocess using LangChain
- Split into meaningful chunks (~500 tokens)
- Display representative chunks
```

Capstone Project Part 4 - Task 1

Document Loading and Preprocessing (RAG Pipeline)

Objective

- Load a document (PDF/TXT/DOCX)
- Preprocess using LangChain
- Split into meaningful chunks (~500 tokens)
- Display representative chunks

```
# 📦 Install required libraries
!pip install langchain langchain-community langchain-text-splitters pymupdf unstructured
```

```
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0
Requirement already satisfied: soupsieve>=1.6.1 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from data
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from datacl
Requirement already satisfied: six>=1.9 in /usr/local/lib/python3.12/dist-packages (from html5lib<2.0.0,>=1.1->
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from html5lib<2.0.0,>=1
Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.12/dist-packages (from langchain-
Requirement already satisfied: packaging<24.0,>=23.2 in /usr/local/lib/python3.12/dist-packages (from langchain-
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<0.2.
Requirement already satisfied: orjson<4.0.0,>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith
Requirement already satisfied: requests-toolbelt<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numb
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic
```

```
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.(
```

```
#  Import required modules
from langchain_community.document_loaders import PyMuPDFLoader, TextLoader, UnstructuredFileLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
```

```
#  Upload your document (PDF/TXT/DOCX)
from google.colab import files
uploaded = files.upload()
```

Choose files Capstone P…b - Colab.pdf

Capstone Project Part 2.ipynb - Colab.pdf(application/pdf) - 607868 bytes, last modified: 24/03/2026 - 100% done

```
Saving Capstone Project Part 2.ipynb - Colab.pdf to Capstone Project Part 2.ipynb - Colab (2).pdf
```

```
#  Load document (auto-detect type)

file_name = list(uploaded.keys())[0]

if file_name.endswith(".pdf"):
    loader = PyMuPDFLoader(file_name)
elif file_name.endswith(".txt"):
    loader = TextLoader(file_name)
else:
    loader = UnstructuredFileLoader(file_name)

documents = loader.load()

print(f"Total pages/documents loaded: {len(documents)}")
```

```
Total pages/documents loaded: 15
```

```
#  Split document into chunks (~500 tokens)

text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=50
)

chunks = text_splitter.split_documents(documents)

print(f"Total chunks created: {len(chunks)}")
```

```
Total chunks created: 58
```

```
#  Display first 3 representative chunks

for i, chunk in enumerate(chunks[:3]):
    print(f"\n--- Chunk {i+1} ---\n")
    print(chunk.page_content)
```

```
--- Chunk 1 ---

Task 1 . Evaluate ViT Image Classification
In this section, we fine-tune a Vision Transformer (ViT) model on the beans dataset,
evaluate its performance, and display the final metrics:
Accuracy
Precision
Recall
F1 Score
Capstone Project Part 2
keyboard_arrow_down
!pip install -q transformers datasets evaluate
!pip install -q --upgrade fsspec
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84.1/84.1 kB 5.2 MB/s eta 0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 202.5/202.5 kB 7.1 MB/s eta

--- Chunk 2 ---

ERROR: pip's dependency resolver does not currently take into account a
datasets 4.0.0 requires fsspec[http]<=2025.3.0,>=2023.1.0, but you have
gcsfs 2025.3.0 requires fsspec==2025.3.0, but you have fsspec 2026.2.0
# Imports
from datasets import load_dataset
import numpy as np
import torch
import evaluate
from transformers import AutoImageProcessor
from datasets import load_dataset
import numpy as np
import torch
import evaluate
```

```
from transformers import (
    AutoImageProcessor,

--- Chunk 3 ---

AutoImageProcessor,
    ViTForImageClassification,
    TrainingArguments,
    Trainer
)
# Load dataset
ds = load_dataset("beans")
ds
24/03/2026, 00:17
Capstone Project Part 2.ipynb - Colab
https://colab.research.google.com/drive/1RPIu4jx-25JMxlq3LMtiQIA-7hXQMan6#scrollTo=xaVoHaw_eXFe&printMode=true
1/15
```

```
# (Optional) Inspect metadata
chunks[0].metadata
```

```
{'source': 'Capstone Project Part 2.ipynb - Colab (2).pdf',
 'file_path': 'Capstone Project Part 2.ipynb - Colab (2).pdf',
 'page': 0,
 'total_pages': 15,
 'format': 'PDF 1.4',
 'title': 'Capstone Project Part 2.ipynb - Colab',
 'author': '',
 'subject': '',
 'keywords': '',
 'creator': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/145.0.0.0 Safari/537.36',
 'producer': 'Skia/PDF m145',
 'creationDate': "D:20260323161730+00'00'",
 'modDate': "D:20260323161730+00'00'",
 'trapped': ''}
```

Capstone Project Part 4 - Task 2

Text Embedding and Vector Store Setup

Objective

- Generate embeddings for text chunks
- Store in vector database (FAISS)
- Perform semantic search
- Retrieve top 3 relevant chunks

```
#  Install additional dependencies
!pip install faiss-cpu sentence-transformers
```

```
Requirement already satisfied: faiss-cpu in /usr/local/lib/python3.12/dist-packages (1.13.2)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
Requirement already satisfied: numpy<3.0,>=1.25.0 in /usr/local/lib/python3.12/dist-packages (from faiss-cpu) (1.
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from faiss-cpu) (23.2)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sente
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transforme
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformer
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.1
Requirement already satisfied: typing_extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from sentence
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (4.67
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->se
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentenc
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sent
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->se
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-tr
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers<6.
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from trans
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers<6.0.0,>=4
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers<6
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->sente
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfa
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->hugging
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->h
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
```

```
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-h
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->hugging
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-h
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggi
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
```

```
from langchain_community.embeddings import HuggingFaceEmbeddings
from langchain_community.vectorstores import FAISS
```

```
#  Create Embedding Model

embedding_model = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/t
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limi
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a
Loading weights: 100%                                    103/103 [00:00<00:00, 355.07it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED   :can be ignored when loading from different task/architecture; not ok if you expect identical arch
```

```
#  Create Vector Store (FAISS)

vector_db = FAISS.from_documents(
    documents=chunks,
    embedding=embedding_model
)

print("Vector database created successfully")
```

```
Vector database created successfully
```

```
#  Perform Semantic Search

query = "What are the key policies or important points discussed in the document?"

results = vector_db.similarity_search(query, k=3)
```

```
#  Perform Semantic Search

query = "What are the key policies or important points discussed in the document?"

results = vector_db.similarity_search(query, k=3)
```

```
#  Display Top 3 Results

print(f"\n Query: {query}\n")

for i, res in enumerate(results):
    print(f"\n--- Result {i+1} ---\n")
    print(res.page_content)
```

```
 Query: What are the key policies or important points discussed in the document?


--- Result 1 ---

24/03/2026, 00:17
Capstone Project Part 2.ipynb - Colab
https://colab.research.google.com/drive/1RPIu4jx-25JMxlq3LMtiQIA-7hXQMan6#scrollTo=xaVoHaw_eXFe&printMode=true
9/15
```

```
--- Result 2 ---

24/03/2026, 00:17
Capstone Project Part 2.ipynb - Colab
https://colab.research.google.com/drive/1RPIu4jx-25JMxlq3LMtiQIA-7hXQMan6#scrollTo=xaVoHaw_eXFe&printMode=true
10/15

--- Result 3 ---

Accuracy : 0.3308
Precision: 0.1103
Recall   : 0.3333
F1 Score : 0.1657
24/03/2026, 00:17
Capstone Project Part 2.ipynb - Colab
https://colab.research.google.com/drive/1RPIu4jx-25JMxlq3LMtiQIA-7hXQMan6#scrollTo=xaVoHaw_eXFe&printMode=true
11/15
```

Capstone Project Part 4 - Task 3

Retrieval-Augmented Generation (RAG) with Conversational Memory

Objective

- Integrate vector store with LLM
- Enable retrieval-based answering
- Add conversational memory
- Simulate multi-turn conversation

```
!pip install openai langchain-openai
```

```
Requirement already satisfied: openai in /usr/local/lib/python3.12/dist-packages (1.109.1)
Requirement already satisfied: langchain-openai in /usr/local/lib/python3.12/dist-packages (0.0.8)
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from openai) (4.12.1)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from openai) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from openai) (0.28.1)
Requirement already satisfied: jiter<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from openai) (0.13.0)
Requirement already satisfied: pydantic<3,>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from openai) (2.12.
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from openai) (1.3.1)
Requirement already satisfied: tqdm>4 in /usr/local/lib/python3.12/dist-packages (from openai) (4.67.3)
Requirement already satisfied: typing-extensions<5,>=4.11 in /usr/local/lib/python3.12/dist-packages (from openai
Requirement already satisfied: langchain-core<0.2.0,>=0.1.27 in /usr/local/lib/python3.12/dist-packages (from lan
Requirement already satisfied: tiktoken<1,>=0.5.2 in /usr/local/lib/python3.12/dist-packages (from langchain-open
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0->openai
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->openai)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->o
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,
Requirement already satisfied: PyYAML>=5.3 in /usr/local/lib/python3.12/dist-packages (from langchain-core<0.2.0,
Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.12/dist-packages (from langchain-co
Requirement already satisfied: langsmith<0.2.0,>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain
Requirement already satisfied: packaging<24.0,>=23.2 in /usr/local/lib/python3.12/dist-packages (from langchain-c
Requirement already satisfied: tenacity<9.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic
Requirement already satisfied: regex>=2022.1.18 in /usr/local/lib/python3.12/dist-packages (from tiktoken<1,>=0.5
Requirement already satisfied: requests>=2.26.0 in /usr/local/lib/python3.12/dist-packages (from tiktoken<1,>=0.5
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0,>=
Requirement already satisfied: orjson<4.0.0,>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<0
Requirement already satisfied: requests-toolbelt<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from l
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests>=2.26.0
```

```
import os
os.environ["OPENAI_API_KEY"] = "xxx"
```

```
# restart session

# !pip uninstall -y langchain langchain-community langchain-openai langchain-core

# !pip install langchain==0.1.20 \
#              langchain-community==0.0.38 \
#              langchain-openai==0.0.8 \
#              faiss-cpu \
#              sentence-transformers
```

```
from langchain_openai import ChatOpenAI
from langchain_community.vectorstores import FAISS

# ✅ Correct memory import
from langchain.memory.buffer import ConversationBufferMemory

# ✅ Correct chain import
from langchain.chains.conversational_retrieval.base import ConversationalRetrievalChain
```

```
# 🤖 Initialize LLM

llm = ChatOpenAI(
    model="gpt-3.5-turbo",   # or gpt-4 if available
    temperature=0
)
```

```
# 🧠 Add Memory

memory = ConversationBufferMemory(
    memory_key="chat_history",
    return_messages=True
)
```

```
qa_chain = ConversationalRetrievalChain.from_llm(
    llm=llm,
    retriever=vector_db.as_retriever(),
    memory=memory
)
```

```
# Turn 1
q1 = "Summarize the document"
r1 = qa_chain({"question": q1})
print("\nUser:", q1)
print("Assistant:", r1["answer"])

# Turn 2
q2 = "Explain it in simple terms"
r2 = qa_chain({"question": q2})
print("\nUser:", q2)
print("Assistant:", r2["answer"])

# Turn 3
q3 = "What are the key takeaways?"
r3 = qa_chain({"question": q3})
print("\nUser:", q3)
print("Assistant:", r3["answer"])
```

```
/usr/local/lib/python3.12/dist-packages/langchain_core/_api/deprecation.py:119: LangChainDeprecationWarning: The
  warn_deprecated(
---------------------------------------------------------------------------
RateLimitError                            Traceback (most recent call last)
/tmp/ipykernel_19809/2713076936.py in <cell line: 0>()
      1 # Turn 1
      2 q1 = "Summarize the document"
----> 3 r1 = qa_chain({"question": q1})
      4 print("\nUser:", q1)
      5 print("Assistant:", r1["answer"])

                              28 frames
/usr/local/lib/python3.12/dist-packages/openai/_base_client.py in request(self, cast_to, options, stream,
stream_cls)
   1045
   1046                 log.debug("Re-raising status error")
-> 1047                 raise self._make_status_error_from_response(err.response) from None
   1048
   1049             break

RateLimitError: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan
and billing details. For more information on this error, read the docs:
https://platform.openai.com/docs/guides/error-codes/api-errors.', 'type': 'insufficient_quota', 'param': None,
```

Next steps: Explain error

Conversational Memory: Enhancing Context-Awareness and Continuity

Conversational memory was implemented using ConversationBufferMemory, which stores the full history of interactions between the user and the assistant.

How Memory Improves Context-Awareness

- Each user query is not treated in isolation
- Previous questions and responses are retained as chat history
- The model uses this history to interpret follow-up queries correctly

Example:

- User: "Summarize the document"
- User: "Explain that in simple terms"

The second query depends on the previous response. Memory allows the model to understand what "that" refers to.

How Memory Enables Continuity

- Maintains a multi-turn conversation flow
- Allows the assistant to:
  - Build on previous answers
  - Avoid repetition
  - Provide more refined responses

Example flow:

1. Initial summary of the document
2. Simplified explanation of the same content
3. Extraction of key takeaways

Each step builds logically on the previous one.

Impact on RAG System

- Combines retrieved knowledge (RAG) with conversation history
- Produces:
  - More relevant answers
  - Better user experience
  - More natural interaction

Key Benefit

Conversational memory transforms the system from a single-turn question-answer model into a context-aware assistant capable of maintaining coherent multi-step interactions.